llmXive Automated Review of 4D Human-Scene Reconstruction from Low-Overlap Captures

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents a coherent methodology and the majority of quantitative claims are well-supported by the provided tables. The numerical values in the ablation studies (Tables 3, 4, 5) and the main results (Tables 1, 2) are internally consistent with the text descriptions.

However, there is a specific discrepancy in the Appendix under the section "GEN3C vs. Ours (Rendering Quality)". The text states: "our rendering on held-out evaluation cameras (PSNR/SSIM/LPIPS: 20.13/0.656/0.215) surpasses GEN3C's raw synthesized views (19.05/0.612/0.329)". These specific numbers (20.13, 0.656, 0.215) do not appear in any of the main tables (Table 1, Table 2) or the ablation tables (Table 6). Table 6 lists "Ours w/o enh." as 20.39/0.657/0.286, which differs from the text's claim. Since the text cites specific metrics that are not present in the data tables, the reader cannot verify this specific comparative claim. The authors should either update the text to match the numbers in Table 6 (or the correct table) or add a new table containing these specific "GEN3C vs. Ours" metrics to ensure the claim is fully supported by the evidence.

Additionally, there is a minor ambiguity in Section 3.1 regarding the generation of human masks for the 481 synthesized views. The text states, "We also obtain human masks for each synthesized view," but the mathematical notation {M_n^1} and the description of the segmentation model suggest masks are only generated for the N input views. If masks are indeed generated for all 481 synthesized views, the method description should clarify how this is achieved (e.g., by propagating masks or re-running the model), as the current text implies a contradiction between the claim and the defined variables.

Action items.

- [**writing**] Appendix Section 'GEN3C vs. Ours (Rendering Quality)' claims specific metrics (PSNR 20.13, SSIM 0.656, LPIPS 0.215) for the proposed method on held-out cameras. These exact numbers do not appear in Table 1, Table 2, or Table 6 (which lists 'Ours w/o enh.' as 20.39/0.657/0.286). The text cites evidence that is not present in the provided tables, making the claim unverifiable. Update the text to match the table data or add a table containing these specific comparison metrics.
- [**writing**] Section 3.1 states 'We also obtain human masks for each synthesized view' to exclude human regions during background optimization. However, the method description and notation {M_n^1} imply masks are only generated for the N input views using a segmentation model. It is unclear how masks are obtained for the 481 synthesized views (e.g., via propagation or re-running the model). Clarify the mask generation process for synthesized views to resolve this logical gap.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`
Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 provides a clear visual overview of the four-stage pipeline with distinct modules and data flow. However, the caption contains a grammatical error ('proposed .') and could better align its text with the specific section labels shown in the diagram.

Figure 2

The figure provides a high-level schematic of the recursive enhancement module but lacks specific implementation details in the diagram itself. The caption is incomplete due to a missing section number reference.

Figure 3

The figure provides a clear schematic of the data flow but lacks explicit visual cues for the recursive loop described in the caption, and the labeling of 'Refined Output' vs 'Enhanced Output' could be more precise to match the text.

Figure 4

Figure 4 effectively presents a qualitative comparison across four scenes (Legoassemble, Grappling, Sword, Karate) and five methods (Dyn-3DGS, MonoFusion, STG, Ours, GT). The grid layout is clear, labels are legible, and the visual evidence strongly supports the caption's claim that the proposed method produces sharper backgrounds and more robust human reconstructions than the baselines.

Figure 5

The figure presents a qualitative comparison but suffers from ambiguous dataset attribution in the caption and a potentially misleading 'GT' row that appears lower quality than the proposed method in some columns, undermining the validity of the comparison.

Figure 6

The figure effectively demonstrates the visual improvement of the recursive enhancement module by comparing raw and enhanced renders. However, the caption contains two instances of missing section numbers ('Sec. .') that need to be filled in.

Figure 7

The figure is visually clear but fails to support its caption's claim of showing results on the EgoExo-4D dataset, as the images appear to be duplicates of scenes shown in other figures (Figure 1 and Figure 5).

Figure 8

The figure effectively demonstrates the visual improvement of background reconstruction after refinement, but the caption contains empty section references that should be corrected.

Figure 9

The figure effectively visualizes the four stages of the enhancement module with clear labels. However, the caption contains a missing section number reference and could be slightly more explicit about the variable definitions.

Figure 10

The figure effectively visualizes the augmented camera trajectory, but the start/end markers in the legend contradict the visual path, and the specific view labels (V0-V3) lack definition.

Figure 11

Figure 11's color-coding scheme is confusing and contradicts its caption, which claims each color represents the same person across cameras while showing multiple people with different colors in the same frame. Additionally, the caption references an incomplete table number.

Figure 12

Figure 12 effectively visualizes the initialized human poses across four distinct scenes using overlaid SMPL meshes on point clouds. The caption clearly explains the content, and the images demonstrate the method's ability to estimate body poses from sparse multi-view inputs without any missing labels or confusing elements.

Action items.

- [**writing**] Figure 1: The caption reads 'Overview of the proposed .' with a missing noun (e.g., 'method' or 'pipeline') after 'proposed'.
- [**writing**] Figure 1: Section references in the diagram (e.g., 'Sec 3.1', 'Sec 3.2') are present, but the caption's stage descriptions do not explicitly map to these section numbers for clarity.
- [**writing**] Figure 2: The caption contains a placeholder '(Sec.)' with a missing section number, failing to cross-reference the detailed text.
- [**science**] Figure 2: The diagram shows a 'Motion-Adaptive Consistency Injection' loop but lacks the specific components mentioned in the related Figure 3 caption (e.g., RAFT, backward flow, EMA), making the mechanism opaque.

- [**science**] Figure 3: The diagram shows 'Refined Output' (O_{t-K}) being warped, but the caption states 'previous enhanced outputs' are warped. While likely referring to the same thing, the diagram should explicitly label the input to the warping step as the 'Enhanced Output' or 'Refined Output' to match the caption's description of the recursive process.
- [**writing**] Figure 3: The ellipsis (...) between the $t-K$ and $t-1$ blocks implies a sequence, but the arrows from the 'Refined Output' blocks go directly into the warping step without showing the recursive loop where the output of one step becomes the input of the next. This makes the 'recursive' nature described in the caption visually ambiguous.
- [**writing**] Figure 5: The caption states 'Dance Xu et al. (Mobile Stage); Yoga Xu et al. (SelfCap)', but the image labels only show 'Tennis', 'Fencing', 'Dance', and 'Yoga' without specifying which dataset corresponds to which column, creating ambiguity about the source of the Tennis and Fencing data.
- [**science**] Figure 5: The 'Ours' row shows significantly cleaner results than the 'GT' (Ground Truth) row in the Tennis and Fencing columns, where 'GT' appears to have motion blur or artifacts not present in the proposed method; this contradicts the expectation that GT should be the cleanest reference and suggests the 'GT' label may be misapplied or the comparison is misleading.
- [**writing**] Figure 6: The caption references 'Sec. .' with a missing section number; please insert the correct section reference.
- [**science**] Figure 7: The caption claims to show qualitative results on EgoExo-4D, but the rendered image is identical to Figure 5 (Dance scene) and Figure 1 (CPR scene), failing to demonstrate the claimed diversity or specific dataset application.
- [**writing**] Figure 8: The caption contains empty section references '(Sec.)' that need to be filled with the correct section numbers.
- [**writing**] Figure 9: The caption contains a broken cross-reference 'Sec. .' where the section number is missing.
- [**writing**] Figure 9: The caption lists 'per-pixel confidence map' as the third item, but the image label uses the variable 'c' without explicitly defining it as a map in the text.
- [**science**] Figure 10: The legend defines a 'Start' (green circle) and 'End' (orange square), but the rendered trajectory shows the orange square at the start of the path and the green circle is not visible, contradicting the legend.
- [**writing**] Figure 10: The labels 'V0', 'V1', 'V2', and 'V3' are present on the plot but are not defined in the legend or the caption.
- [**science**] Figure 11: The caption claims 'Each color represents the same person matched across 4 cameras', but the image shows multiple people (Person A, B, C) simultaneously in the same frame, each with different colored boxes. This contradicts the caption's explanation of the color coding scheme.

- [**writing**] Figure 11: The caption references 'Table .' with a missing table number, making it impossible to verify the claimed 97.8% association accuracy.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured and uses standard terminology for the computer vision and graphics community (e.g., 3DGS, SMPL, NeRF, LBS). However, a competent reader from an adjacent field (e.g., a computer vision researcher not specializing in 4D reconstruction or a graphics researcher new to diffusion-based refinement) would encounter several undefined terms and symbols that create minor barriers to full comprehension.

Specifically, the introduction of "SAM3" in Section 3.1 assumes knowledge of a specific model variant that is not universally standard like "SAM." Similarly, several mathematical symbols in the Method section (σ_p, σ_θ, $\mathcal{B}$, K) are introduced in equations without immediate textual definition, forcing the reader to infer their meaning from context or search for them later. While the context often makes the meaning clear to an insider, the lack of explicit definition violates the self-contained requirement for an adjacent-field PhD.

Additionally, the metric "Warp-L2" is used in the ablation study without a precise operational definition of how the error is aggregated or normalized. These are not fundamental flaws in the science, but they represent "exclusion by omission" where the authors have not taken the extra step to ensure their notation is fully accessible to a smart reader outside their immediate subfield. Addressing these by adding brief parenthetical definitions or one-sentence glosses at the point of first use would resolve these issues.

Action items.

- [**writing**] Section 3.1 (Preprocessing): The term 'SAM3' is used without definition. While 'SAM' (Segment Anything Model) is standard, 'SAM3' appears to be a specific variant or the authors' own nomenclature (referenced as carion2025sam). Define it at first use, e.g., 'SAM3 (Segment Anything Model 3, a concept-based segmentation model)...'.
- [**writing**] Section 3.2 (Cross-View Identity Association): The symbol σ_p and σ_heta appear in Equation 1 without definition. While context suggests they are scale parameters for the Gaussian kernels, explicitly define them in the text immediately following the equation (e.g., 'where σ_p and σ_heta are the spatial and pose scale parameters, respectively').
- [**writing**] Section 3.3 (3D Pose Triangulation): Equation 3 introduces $\mathcal{B}$ as 'the set of reliable bone pairs' but does not define the criteria for 'reliable' or how this set is determined algorithmically. Add a brief clause explaining the selection criterion (e.g., 'where $\mathcal{B}$ is the set of bone pairs with reprojection error below threshold X').
- [**writing**] Section 3.4 (Human Reconstruction): The term 'canonical pose' is used to describe

the reference state for Gaussians. While standard in SMPL-based reconstruction, explicitly define it for adjacent-field readers as 'a standard T-pose or A-pose reference configuration' to ensure clarity.

- **[writing]** Section 3.5 (Recursive Enhancement Module): The variable K in Equation 5 is used as the number of previous frames but is not defined in the text preceding the equation. Define it explicitly: 'where K is the number of previous frames considered for injection (set to 3 in our experiments).'
- **[writing]** Section 4.1 (Setup): The metric 'Warp-L2' is introduced in Table 3 and the text without a definition of the specific error calculation (e.g., L2 norm of RGB difference after warping). Define it briefly: 'Warp-L2 measures the L2 error between consecutive frames after optical flow warping, computed as...'

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent argument for decoupling background and human reconstruction in low-overlap settings. The logical flow from the problem definition (sparse views, entangled errors) to the proposed solution (diffusion for background, SMPL for humans) is sound. The premises regarding the limitations of existing methods are consistently used to motivate the specific components of the pipeline.

However, there is a minor logical disconnect in the Ablation Study (Section 4.3). The text states: "removing the attention-gating module causes the largest performance drop," yet the corresponding table (Table 5, `tab:ablation_inject`) compares "w/o Injection" against the full method. The table shows that removing this component results in *identical* PSNR and SSIM scores, with the only significant change being in Warp-L2 (temporal consistency) and a slight LPIPS increase. The text's claim of a "largest performance drop" in general metrics is not supported by the data in the table, which shows the drop is specific to temporal consistency and perceptual quality, not reconstruction fidelity. This is a non-entailed conclusion where the text overgeneralizes the impact of the ablated component.

Additionally, there is a slight scope ambiguity between the Abstract and the Results. The Abstract claims SOTA results on "four real-world datasets," while the detailed analysis in Section 4.3 and Table 5 explicitly references "6 360-degree scenes." While not a fatal contradiction, the phrasing in the Abstract could be tightened to ensure the reader understands the specific scope of the quantitative claims.

Overall, the argument holds together well, but the specific interpretation of the ablation results in the text requires alignment with the table data to avoid misleading the reader about the magnitude of the performance drop.

Action items.

- [**writing**] Section 4.3 claims 'removing the attention-gating module causes the largest performance drop,' but Table 5 shows PSNR/SSIM are identical with/without injection. The text overgeneralizes the impact; clarify that the drop is specific to temporal consistency (Warp-L2) and LPIPS, not static fidelity.
- [**writing**] The Abstract claims SOTA on 'four real-world datasets,' while Section 4.3 details results on '6 360-degree scenes.' Ensure the Abstract's scope matches the specific scene breakdown in the results to avoid implying a broader evaluation than presented.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper presents a robust methodology for 4D reconstruction under sparse-view constraints, but the rhetoric in the abstract and conclusion occasionally exceeds the specific boundaries of the experimental evidence.

First, the term "in-the-wild studio capture" is used to frame the problem and the solution's applicability. While the introduction defines this as "sparse, low-overlap cameras... in sports venues, healthcare facilities, and home environments," the experimental validation is restricted to four specific, curated datasets (EgoHumans, Harmony4D, Mobile Stage, SelfCap) with synchronized, calibrated inputs. The phrase "in-the-wild" typically connotes uncalibrated, asynchronous, and highly variable real-world data. By claiming the method solves the "in-the-wild" problem based solely on these curated benchmarks, the paper risks overgeneralizing its robustness to truly unstructured environments. The abstract's claim of achieving state-of-the-art results across "diverse real-world datasets" should be tempered to reflect the specific nature of the evaluated datasets.

Second, the conclusion asserts that the method brings high-fidelity capture "closer to practical deployment." However, the limitations section explicitly notes that the method cannot reconstruct dynamic objects (like basketballs or props) and that shadows are baked into the static background, failing to follow human motion. These are significant functional gaps for many "practical" use cases in sports or interactive scenarios. The claim of practical readiness overreaches the demonstrated capabilities; the text should qualify the deployment claim to exclude scenarios requiring dynamic object handling or accurate shadow interaction.

Finally, the claim of being the "first pipeline" to achieve this specific combination of high-fidelity reconstruction and consistent refinement is strong. While the paper compares against relevant baselines, the absolute "first" claim requires a comprehensive survey of all prior art to be fully substantiated. A more precise phrasing, such as "the first among Gaussian-based pipelines" or "to our knowledge, the first to combine these specific components," would better align the claim with the evidence provided.

These are primarily rhetorical adjustments to ensure the scope of the claims matches the scope of the evidence, rather than fundamental flaws in the science.

Action items.

- [**writing**] Abstract claims 'in-the-wild' success, but experiments use 4 curated, synchronized datasets. Narrow claim to 'evaluated sparse-view benchmarks' or define 'in-the-wild' scope explicitly.
- [**writing**] Conclusion claims 'practical deployment' readiness, yet limitations admit failure on dynamic objects and shadows. Qualify deployment claim to exclude scenarios requiring these features.
- [**writing**] Abstract claims 'first pipeline' without qualifying scope. Add 'to our knowledge' or specify 'among Gaussian-based pipelines' to align with the limited baseline comparison provided.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a computer vision method for 4D human-scene reconstruction using public datasets (EgoHumans, Harmony4D, Mobile Stage, SelfCap) and pre-trained models. The work does not involve the collection of new human-subject data, nor does it release any datasets containing Personally Identifiable Information (PII). The authors explicitly state the use of public benchmarks and standard pre-trained models (e.g., SAM, GEN3C, CoMotion) for segmentation, view synthesis, and pose estimation.

There are no indications of dual-use capabilities that lower the barrier to specific harms (e.g., automated surveillance, deepfake generation for deception, or cyber-attacks) beyond the general capabilities of the underlying generative models, which are not the primary contribution of this work. The paper does not describe operational details for exploiting vulnerabilities or synthesizing hazardous materials. Furthermore, the authors acknowledge limitations regarding dynamic objects and shadows, demonstrating appropriate scientific transparency.

As this is a third-party preprint, the absence of an explicit IRB statement is not a violation, as the research relies entirely on existing public datasets and does not involve new human subject interactions. The paper does not raise foreseeable, non-trivial safety or ethical risks that require mitigation or disclosure beyond what is currently present. The verdict is `accept` with no action items.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling pipeline for 4D reconstruction in low-overlap settings, but the experimental design in the ablation studies and main comparisons leaves open the possibility that the reported gains stem from data volume rather than the proposed architectural innovations.

First, the quantitative results in Tables 1 and 2 are reported as single-point estimates without any measure of variance (standard deviation or confidence intervals). In 3D reconstruction tasks, performance can be highly sensitive to random initialization, camera pose estimation noise, and diffusion model stochasticity. A 2.41 PSNR gain (Table 2) is substantial, but without reporting results across multiple random seeds (e.g., 3-5 runs), it is impossible to distinguish a robust methodological improvement from a lucky initialization or a specific favorable scene configuration. The claim of "state-of-the-art" results requires demonstrating that this margin holds consistently across stochastic variations.

Second, the comparison against baselines in Section 4.1 introduces a significant confound regarding data volume. The proposed method leverages 481 synthesized novel views generated by a video diffusion model to train the background Gaussians. In contrast, the baselines (Dyn-3DGS, MonoFusion, STG) are trained only on the 4 original sparse input views. The observed performance gap could be entirely attributed to the massive increase in supervision data (120x more views) rather than the specific "decoupled" architecture or the "motion-adaptive consistency injection." To isolate the contribution of the proposed method, the authors must run a control experiment where the baselines are also trained on the same set of 481 synthesized views (or a matched subset). If the baselines still underperform with the same data, the architectural contribution is validated; if they close the gap, the primary driver is the data augmentation, not the method.

Finally, the ablation on motion-adaptive consistency injection (Table 3) presents a trade-off that is not fully quantified. The injection reduces Warp-L2 (a proxy for temporal consistency) by 23% but slightly degrades LPIPS and leaves PSNR/SSIM unchanged. The authors argue this improves "viewing experience" by eliminating flickering, but this is currently supported only by qualitative figures (Figure 6). To support this claim scientifically, a quantitative metric for temporal consistency (such as temporal L1 error between consecutive frames or a flow-based consistency score) should be reported on the final enhanced output. Without this, the trade-off remains an unverified assertion.

Action items.

- **[writing]** The paper presents a compelling pipeline for 4D reconstruction in low-overlap settings, but the experimental design in the ablation studies and main comparisons leaves open the possibility that the reported gains stem from data volume rather than the proposed architectural innovations. First, the quantitative results in Tables 1 and 2 are reported as single-point estimates without any measure of variance (standard deviation or confidence intervals). In 3D reconstruction tasks, performance can be

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this paper is generally consistent with common practices in the computer vision and graphics community (reporting mean metrics on fixed datasets), but it lacks the rigor required to distinguish genuine improvements from random variance, particularly given the small number of test scenes.

The primary issue is the absence of uncertainty quantification. Tables 1 and 2 present performance metrics (PSNR, SSIM, LPIPS) to two decimal places (e.g., 18.58, 0.569) as if they were exact constants. In stochastic deep learning pipelines involving Gaussian Splatting and diffusion models, results typically exhibit run-to-run variance. Without reporting the standard deviation (SD) or standard error (SE) across multiple random seeds (e.g., 3–5 runs), the reader cannot determine if the reported improvements (e.g., +2.4 PSNR) are statistically significant or within the noise floor of the training process. The field norm for such claims is "mean ± SD over N seeds."

Similarly, Table 3 (ablation_association) reports a single aggregate accuracy of 97.8% across 8 scenes. With such a small sample size (N=8), the mean is highly sensitive to outliers. A single scene with poor association would significantly lower the average. The authors should provide the per-scene accuracy values or the standard deviation across the 8 scenes to demonstrate the robustness of the association strategy.

Furthermore, the claim of "State-of-the-art" results is based on 18 pairwise comparisons (6 scenes × 3 metrics) without addressing the multiple comparisons problem. Highlighting the best result in every cell without a formal hypothesis test (e.g., paired t-test) and correction (e.g., Holm-Bonferroni or Benjamini-Hochberg) inflates the Type I error rate. While the magnitude of the improvement in some cases (e.g., LPIPS) appears large, the statistical framework to support the "significant" nature of these wins is missing. The authors should either run the appropriate statistical tests with correction or soften the language to describe the results as "empirically superior" without invoking statistical significance.

Finally, the robustness analysis in Table 5 (sensitivity to input noise) reports PSNR drops to two decimal places (e.g., 20.04) but does not indicate if these values are averages over seeds or single runs. Given the sensitivity analysis involves re-training, the variance across seeds should be reported to ensure the observed robustness is not an artifact of a lucky initialization.

Action items.

- [**writing**] The statistical reporting in this paper is generally consistent with common practices in the computer vision and graphics community (reporting mean metrics on fixed datasets), but it lacks the rigor required to distinguish genuine improvements from random variance, particularly given the small number of test scenes. The primary issue is the absence of uncertainty quantification. Tables 1 and 2 present performance metrics (PSNR, SSIM, LPIPS) to two decimal places (e.g., 18.58, 0.569) as if they w

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured, with a clear logical progression from problem definition to method and evaluation. However, several sentences suffer from complex nesting or ambiguous referents that force the reader to re-parse the text to recover the intended meaning.

In Section 3.1, the description of mask usage contains a relative clause ambiguity ("which are used...") that could refer to the views or the masks. While context suggests the latter, the syntax is imprecise. Similarly, Section 3.2 combines the initial cross-view association logic with the temporal re-assignment logic in a single paragraph, creating a slight cognitive load as the reader tracks two distinct temporal mechanisms. Splitting these would clarify the distinct roles of the Hungarian algorithm (spatial) and the re-assignment logic (temporal).

The most significant readability friction occurs in Section 3.4. The explanation of the motion-adaptive consistency injection relies on a long, nested sentence ("Inspired by..., we observe that since..., injecting... is equivalent to..."). The "since" clause interrupts the flow between the observation and the conclusion. A rewrite that separates the observation from the justification would significantly improve clarity.

Finally, the ablation study in Section 4.2 occasionally jumps between describing the method, presenting the result, and interpreting the trade-off within the same paragraph. A stricter adherence to a "Problem -> Method -> Result -> Interpretation" structure in these paragraphs would make the experimental narrative more linear and easier to follow. These are minor structural and syntactic issues that, once addressed, will allow the reader to move through the technical details without friction.

Action items.

- [**writing**] The paper is generally well-structured, with a clear logical progression from problem definition to method and evaluation. However, several sentences suffer from complex nesting or ambiguous referents that force the reader to re-parse the text to recover the intended meaning. In Section 3.1, the description of mask usage contains a relative clause ambiguity ("which are used...") that could refer to the views or the masks. While context suggests the latter, the syntax is imprecise. Similarly, Sec